

Machine Learning and Software Engineer with 10+ years of experience across machine learning model training and deployment, and backend architecture (Python, Java/Spring), covering the implementation of machine learning projects from concept and requirements gathering to data preparation and production monitoring.

EMPLOYMENT

Sr. ML Engineer

Wiley

Summer 2019-Present

Manuscript AI analyzer (May 2019 – Present)

The service extracts meta-information like authors, affiliations, funders and others from a scientific manuscript to make its submission easier and faster. The project processes tens of thousands of papers from thousands of journals of different publishers including Wiley and IEEE every day in a few seconds each running entirely on CPUs.

- Developed a NER service to extract institutional data like department, institution etc. from text affiliations, improving Ringgold Database search and institution disambiguation. The service processes over 2 million requests per month running on a single CPU within a Kubernetes cluster, achieving an F1 score of over 0.97 for specific entities.
- Trained models and developed a service that extracts manuscript citations from the main text and tables. The service identifies citations that do not have references. A fast Spacy model for scanning main text and a transformer-based model with axial attention for scanning tables were used.
- Trained and implemented a shallow binary classification model to identify corresponding authors.
- Simplified funders-grants extraction and relation models. A complex pipeline of a sentence segmentation model, sentence classifier, NER and Dependency Parser was replaced with Span Categorizer.
- Upgraded an author extraction pipeline by migrating from NER to a Span Categorizer and introducing the extraction of academic degrees and honorifics.
- Improved a paragraph classification model by adding additional labels like bio sections and subheadings. Resolved the lack of training data by employing approaches like text retrieval and an LLM as a judge.
- Tested and integrated the project with a PDF recognition service that improved the recognition of text order and block and in-text formulas.
- Prepared training and validation data for different models using approaches like fuzzy matching, text retrieval, LLM data augmentation.
- Developed and integrated monitoring and metrics explorer tools for the models which help to evaluate data drift in production.
- Implemented a feedback pipeline to pull new training data from a production environment.
- Implemented a CI/CD pipeline that automated building and deployment of the service.
- Introduced MLflow in the project to make ML experiments more transparent and reproducible.

Software engineer ->

Sr. software engineer

Wiley

Winter 2017 – Spring 2019

Reviewer finder and journal finder services (October 2017 – April 2019)

- Developed REST services for searching relevant journals and reviewers based on the published articles information and taxonomy terms using Elasticsearch and TF-IDF algorithm; integrated them with Jenkins; developed automation tests and automated the deployment to environments.

Classification tool (May 2017 – September 2017)

Designed and implemented from scratch a classification service as a full-stack developer in ReactJS/Java.

Delivery platform (February 2017 – September 2017)

- Developed from scratch a clean-up service to remove obsolete data.
- Implemented business logic and authentication/authorization for delivering article classifications to the Literaturum system.
- Wrote performance, functional, integration and end-to-end tests.

Software Developer	Virgin Connect	Spring 2016 – Winter 2017
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Managed all Java applications for the Moscow Region division running in production, ensuring high availability and performance.

Integration with document management system Diadoc (September 2016 – February 2017)

- Developed a part of logic to export data from Microsoft SQL Server to xml and pdf files speeding up document processing by eliminating manual work.

Billing admin interface (May 2016 – October 2016)

- Implemented a billing admin interface as a full-stack developer.

RADIUS authorization service (May 2016 – October 2016)

- Developed an interaction and synchronization logic between the application and user info databases for authentication. Spring framework, Oracle DB as a primary DB with PostgreSQL as a fail-safe DB and Docker were used.
- Implemented Python scripts for parsing RADIUS logs and store entries in Oracle Warehouse.

Software Developer	Smile Moscow Region ISP	Spring 2015 – Spring 2016
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Payment system (March 2015 – May 2016)

- Designed and implemented a fault-tolerant payment system from scratch using the Spring framework, working collaboratively within an Agile/Scrum team. Developed the logic to support payments from banks.
- Developed Python scripts for Nagios to monitor the status of the application servers.
- Backed up the payment system's servers.

User information message (October 2015 – January 2016)

- Full refactoring of legacy code (separated business logic and presentation; integrated project with Maven)
- Developed and introduced a new user interface using Bootstrap framework.

User personal area (August 2015 – December 2015)

- Set up a frontend nginx server.
- Automated deployment process on dev, test and production servers using Ansible.

Partner feedback web application (September 2015 – November 2015)

- Enhanced the quality of legacy code and adopted Maven automatic build and deploy.
- Integrated partners' requests with Redmine Issue tracker system using REST API.

Intern consultant -> Jr.

software developer	IBS	Fall 2013 – Spring 2015
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Enterprise portal development (November 2013 – March 2015)

- Developed a web application for displaying and editing transport information using ExtJs and Liferay Portal.
- Designed and implemented a gallery component for displaying photos and videos using pure JS and SAP Java stack.
- Wrote a portal component for conducting polls using SAP Java Stack.
- Wrote a web service to update spreadsheets using SOAP and SAP Java stack.

Automation engineer	EMICON	Winter 2012 – Fall 2013
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Software system to automate the creation of design documentation (January 2012 – October 2013)

Designed, coded, tested, introduced and maintained a CAD software system to automate the creation of design documentation of programmable logic controllers which reduced the time of their development by 30%.

EDUCATION

Moscow, Russia	Bauman Moscow State Technical University	Fall 2007 – Fall 2014
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Bachelor's and Master's degree with honors in Computer systems and networks, June 2013. GPA: 4.0. PhD in computer science: not completed.

Coursework: Programming Languages and algorithms; Databases; Circuit engineering; Computer networks; Computer systems; Microprocessor Systems; Signal Processing.

Diploma thesis: "Automatic fire suppression system of an acceptance station at an oil refinery plant".

Online courses	Fall 2013 – Present
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[Sedgewick's Algorithms Course](#); [Web Applications Architecture](#); [Stanford's Machine Learning](#);

CERTIFICATES

English	IELTS: 8.0/C1 (L: 8.5, R: 8.5, W: 8.0, S: 7.0)	March 2019
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ADDITIONAL TECHNICAL EXPERIENCE

Study projects

- [Titanic prediction](#) (Fall 2019) – the titanic survival problem model with 80.3% accuracy on the test set
- [MacOS environment](#) (Spring 2019) – Ansible playbook to automatically deploy developer's environment
- [Key-value cache 3](#) (Winter 2018) – designed and implemented API for a thread-safe key-value cache concept
- [Key-value cache 2](#) (Summer 2018) – designed and implemented API for a key-value cache concept
- [Todo](#) (Spring 2017 – Winter 2017) – a web application for getting things done concept. Written in Java (server) and JS (client) with the usage of Spring (Boot, MVC, Security, Data), Hibernate, Maven, Travis CI; React, npm, webpack; Docker
- [Powerlifters DB Interface](#) (Summer 2016 – Autumn 2016) – a web application that shows different ways to connect to a PostgreSQL DB from Java and that can be launched in Docker.
- [Game server](#) (Summer 2014) – implementation of a game server written in Java. This project is developed with usage of: Jetty HTTP server, FreeMarker Java Template Engine, Selenium, JDBC, Hibernate.
- [Slideshow gallery](#) (Spring 2014) – simple gallery written in pure JS.
- [Game XO](#) (Winter 2013) – Implemented an AI (Minimax) algorithm for an Android game TicTacToe.

SKILLS, LANGUAGES AND TECHNOLOGIES

Main:

- Python, PyTorch, Spacy, scikit-learn.
- Java, Spring (MVC, Data, Boot, Security), Hibernate.

Additional:

- Linux.
- GCP; Docker, k8s; Github Actions, Jenkins; Ansible.
- Tensorflow, MLflow.
- SQL: MySQL, PostgreSQL; Elasticsearch.
- Apache Kafka.
- Tomcat; Nginx.

Also have experience with:

- JS, React.
- AWS.